Jueun **Kim**

Ph.D. Student in CS, POSTECH

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EDUCATION

Ph.D., Computer Science and Engineering *Sep 2025 – Present*

Pohang University of Science and Technology (POSTECH), Pohang, South Korea

- Advisor: Prof. Wook-Shin Han

B.S., Computer Science and Engineering *Feb 2021 – Aug 2025*

Pohang University of Science and Technology (POSTECH), Pohang, South Korea

- Ranked 1st in the department
- Graduated Summa Cum Laude
- GPA: 4.04/4.3

Exchange Student, Computer Science *Sep 2024 – Feb 2025*

Technical University of Munich (TUM), Munich, Germany

- Achieved top grades in Deep Learning and NLP coursework

RESEARCH INTERESTS

Multi-hop and multi-modal information retrieval; retrieval for task-performing agents; long-context management; agent memory

EXPERIENCE

Intern *Jul 2024 – Aug 2024*

SK hynix

- Mobile FTL Team — developed an AI model predicting user commands to improve flash translation layer performance

Student Intern *Jun 2023 – Aug 2025*

Data Systems Lab, POSTECH

- Research Project I: proposed the Text-DB QA task and built its benchmark (ICLR 2026)
- Research Project II: developed an open-domain text-table QA model

PUBLICATIONS

CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG *arXiv preprint arXiv:2610.00923*

Hyojeong Yun*, **Jueun Kim***, Wook-Shin Han (*equal contribution)

Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering *NeurIPS 2026*

Jueun Kim, Sungho Park, Wook-Shin Han

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *ICLR 2026*

Sungho Park, **Jueun Kim**, Wook-Shin Han

PROJECTS

CANOPY: Adaptive-Granularity Evidence Compression for Multimodal RAG *2026*

arXiv preprint arXiv:2610.00923

Represents retrieved items as hierarchies and selects evidence at several granularities at once, since choosing a retrieval granularity does not settle how much of each item to keep — cut reader-input evidence tokens by 14.2–27.7% at comparable accuracy across five QA benchmarks over a 33M-item heterogeneous corpus.

Hi-Q: Hierarchical Evidence-guided Query Refinement for Multi-Hop Question Answering *2026*

NeurIPS 2026

Grows a query tree whose shape is decided by whether retrieved evidence already supports each unit, rather than by a fixed decomposition template or a pre-built graph — 52.3 EM / 64.0 F1 averaged over three multi-hop QA benchmarks under full-corpus retrieval, 15.1 EM over IRCoT.

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *2026*

ICLR 2026

Automates table-text multi-hop benchmark construction by unifying tables and text into one relational representation and generating nested queries at a target hop count — thousands of QA pairs at a quarter of HybridQA's annotation time, on which models scoring above 70 F1 on HybridQA lose more than 30 F1.

Home AI Agent with Event Memory *2026*

Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)

Merges continuing home-camera scenes into events while preserving observation counts and state transitions, so earlier situations can be recovered without replaying footage — 72 observations into 6 events (91.7% fewer list entries), citing the designated event in 6 of 7 past-event questions.

Life-Event-Based Public Service AI Agent *2026*

Korean government-hosted AI competition — finalist (top 20 of 1,100+ teams)

Keeps the user's conditions and completion state as persistent memory and uses it to decide what to surface next across procedures spanning multiple steps and deadlines — 806 procedure items across 50 life events linked by 639 precedence relations, cutting precedence violations from 242 under deadline-only ordering to 0 on a 500-case regression.

User Command Prediction for Mobile Flash Translation Layers *2024*

SK hynix internship — Grand Prize

Predicts the next user command from observed workload patterns so the flash translation layer can prepare for it instead of reacting once the request arrives.

TALKS

SPARTA: Scalable and Principled Benchmark of Tree-Structured Multi-hop QA over Text and Tables *Jul 2026*

2026 Global AI Frontier Lab Workshop, Seoul

HONORS & AWARDS

POSTECHIAN Fellowship (Creative) *Nov 2025*

POSTECH

Grand Prize, SK hynix Internship Program *Aug 2024*

SK hynix

Popular Award, Undergraduate Research Project II *Jun 2024*

POSTECH Computer Science and Engineering Department

National Science & Technology Scholarship *Aug 2023*

KOSAF (Korea Student Aid Foundation)

Global Leadership Program Scholarship *Sep 2022*

POSTECH Computer Science and Engineering Department

TEACHING

Teaching Assistant — AI Application Specialist Course *2026*

Samsung Electronics

Teaching Assistant — Introduction to Data Analysis *2026*

POSTECH

Lecturer — POSTECHx Lecture, Pintos *2024*

POSTECH

Teaching Assistant — Data Programming Course *2023*

Samsung Electronics